

Policy Optimization

Suppose we aim to find an optimal policy π^* , i.e.,

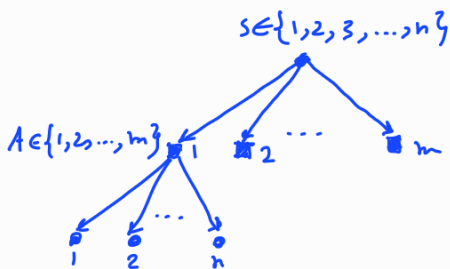
$$\begin{aligned} & V^{\pi^*}(s) \geq V^{\pi}(s) \quad \forall s \in S, \forall \pi \in \Pi \\ \text{or} \quad & V^{\pi^*} \geq V^{\pi} \quad \forall \pi \in \Pi \\ \text{or} \quad & V^{\pi^*} = V^* = \max_{\pi \in \Pi} V^{\pi} \end{aligned}$$

Bellman Optimality Equations:

The optimal value functions, i.e., V^* and Q^* , satisfy the following equations:

$$V^*(s) = \max_{a \in A} \left[\overbrace{R(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} [V^*(s')]}^{Q^*(s,a)} \right] \quad \forall s \in S$$

$$Q^*(s,a) = R(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} \left[\overbrace{\max_{a' \in A} Q^*(s',a')}^{V^*(s')} \right] \quad \begin{matrix} \forall s \in S \\ \forall a \in A \end{matrix}$$



$$V^{\pi}(s) = \mathbb{E}_{a \sim \pi} [R(s,a) + \gamma \mathbb{E}_{s'} [V^{\pi}(s')]] \quad \text{Bellman consistency equation}$$

$$V^* = \max_{a \in A} [R(s,a) + \gamma \mathbb{E}_{s'} [V^*(s')]] \quad \text{Bellman optimality equation}$$

Theorem [Bellman optimality]: π^* is an optimal policy if and only if v^{π^*} satisfies the Bellman optimality equation.

Proof (sketch):

① π^* an optimal policy $\Rightarrow v^{\pi^*}$ satisfies the Bellman optimality equation

Consider the following deterministic, stationary policy

$$\tilde{\pi}(s) = \arg \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [v^{\pi^*}(s')]] \quad \forall s \in S$$

One can show that $v^{\pi^*} = v^{\tilde{\pi}}$ and $v^{\tilde{\pi}}$ satisfies the Bellman optimality equation.

② v^{π} satisfies the Bellman optimality equation $\Rightarrow \pi$ an optimal policy

Consider an optimal policy π^* .

By ①, we know that v^{π^*} satisfies the Bellman optimality equation.

One can show that $|v^{\pi}(s) - v^{\pi^*}(s)| \leq 0$ for all $s \in S$.

proof:

① π^* an optimal policy $\Rightarrow v^{\pi^*}$ satisfies the Bellman optimality equation

Consider the following deterministic, stationary policy

$$\tilde{\pi}(s) = \arg \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi^*}(s')]] \quad \forall s \in S$$

We want to show that $v^{\pi^*} = v^{\tilde{\pi}}$ and $v^{\tilde{\pi}}$ satisfies the Bellman optimality equation.

$$\begin{aligned} v^{\pi^*}(s) &= \mathbb{E}_{a \sim \pi^*(s)} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi^*}(s')]] && \text{Bellman consistency equation} \\ &\leq \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi^*}(s')]] && \text{arg} \leq \max \\ &= R(s, \tilde{\pi}(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \tilde{\pi}(s))} [v^{\pi^*}(s')] && \text{definition of } \tilde{\pi} \end{aligned}$$

$$\text{In vector form: } v^{\pi^*} \leq R^{\tilde{\pi}} + \gamma P^{\tilde{\pi}} v^{\pi^*} \quad \xrightarrow{\text{subtract } v^{\tilde{\pi}}}$$

$$\begin{aligned} \underline{\underline{v^{\pi^*} - v^{\tilde{\pi}}}} &\leq R^{\tilde{\pi}} + \gamma P^{\tilde{\pi}} v^{\pi^*} - v^{\tilde{\pi}} \\ &= R^{\tilde{\pi}} + \gamma P^{\tilde{\pi}} v^{\pi^*} - (\underline{R^{\tilde{\pi}} + \gamma P^{\tilde{\pi}} v^{\tilde{\pi}}}) \\ &= \gamma P^{\tilde{\pi}} (\underline{\underline{v^{\pi^*} - v^{\tilde{\pi}}}}) \end{aligned}$$

repeat the argument

$$\leq \gamma P^{\tilde{\pi}} (\gamma P^{\tilde{\pi}} (v^{\pi^*} - v^{\tilde{\pi}})) = \gamma^2 (P^{\tilde{\pi}})^2 (v^{\pi^*} - v^{\tilde{\pi}})$$

$$\vdots$$
$$\leq \underbrace{\gamma^K (P^{\tilde{\pi}})^K}_{\text{non stochastic}} (v^{\pi^*} - v^{\tilde{\pi}})$$

$$\text{let } K \longrightarrow \infty \quad \underset{\gamma < 1}{\Rightarrow} \quad v^{\pi^*} - v^{\tilde{\pi}} \leq 0 \quad \Rightarrow \quad v^{\pi^*} \leq v^{\tilde{\pi}}$$

On the other hand, π^* is an optimal policy $\Rightarrow v^{\pi^*} \geq v^{\tilde{\pi}}$

$$\Rightarrow v^{\pi^*} = v^{\tilde{\pi}}$$

Also,

$$v^{\tilde{\pi}}(s) = E_{a \sim \tilde{\pi}(s)} [R(s, a) + \gamma E_{s' \sim P(\cdot | s, a)} [v^{\tilde{\pi}}(s')]]$$

definition of $\tilde{\pi}$

$$\stackrel{\text{definition of } \tilde{\pi}}{=} \max_{a \in A} [R(s, a) + \gamma E_{s' \sim P(\cdot | s, a)} [v^{\tilde{\pi}}(s')]]$$

$\Rightarrow v^{\tilde{\pi}}$ satisfies the Bellman optimality equation.

$\Rightarrow v^{\pi^*}$ satisfies the Bellman optimality equation.

② v^π satisfies the Bellman optimality equation $\Rightarrow \pi$ an optimal policy

Consider an optimal policy π^* .

By ①, we know that v^{π^*} satisfies the Bellman optimality equation.

We want to bound $|v^\pi(s) - v^{\pi^*}(s)|$ for all $s \in S$.

$$|v^\pi(s) - v^{\pi^*}(s)| \stackrel{\text{assumption} + \text{①}}{=} \left| \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^\pi(s')] \right] - \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi^*}(s')] \right] \right|$$

$$\begin{aligned} & \left| \max_x g_1(x) - \max_x g_2(x) \right| \leq \max_x |g_1(x) - g_2(x)| \\ & \leq \max_{a \in A} \left| \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^\pi(s')] \right] - \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi^*}(s')] \right] \right| \end{aligned}$$

$$\stackrel{\text{linearity of expectation}}{=} \max_{a \in A} \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^\pi(s') - v^{\pi^*}(s')] \right|$$

$$\stackrel{\text{Jensen's inequality}}{\leq} \max_{a \in A} \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [|v^\pi(s') - v^{\pi^*}(s')|]$$

$$\stackrel{\text{repeat the argument}}{\leq} \max_{a \in A} \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in A} \gamma \mathbb{E}_{s'' \sim P(\cdot | s', a')} [|v^\pi(s'') - v^{\pi^*}(s'')|] \right]$$

joint maximization

sequential maximization $\Rightarrow$

$$\leq \gamma^2 \max_{\substack{a \in A \\ a' \in A}} \mathbb{E}_{\substack{s' \sim P(\cdot | s, a) \\ s'' \sim P(\cdot | s', a')}} [|V^\pi(s') - V^\pi(s'')|]$$

$\vdots$

$$\leq \gamma^k \max_{\substack{a_0 \in A \\ \vdots \\ a_{k-1} \in A}} \mathbb{E}_{\substack{s_1 \sim P(\cdot | s, a_1) \\ \vdots \\ s_k \sim P(\cdot | s_{k-1}, a_{k-1})}} [|V^\pi(s_k) - V^{\pi^*}(s_k)|]$$

bounded if $R(\cdot, \cdot)$ bounded & $\gamma < 1$

Let $k \rightarrow \infty \quad \gamma < 1 \Rightarrow |V^\pi(s) - V^{\pi^*}(s)| \leq 0 \quad \forall s \in S$

$V^{\pi^*}(s) \geq V^\pi(s)$

$\Rightarrow V^\pi(s) = V^{\pi^*}(s) \quad \forall s \in S$

$\Rightarrow \pi$ is an optimal policy



Corollary: The deterministic stationary policy

$$\pi(s) = \operatorname{argmax}_{a \in A} \left[R(s, a) + \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^*(s')]}_{Q^*(s, a)} \right] \quad \forall s \in S$$

or $\pi(s) = \operatorname{argmax}_{a \in A} Q^*(s, a)$

is an optimal policy.

Value Iteration

VI aims to find (approximate) the optimal value function v^* through the fixed-point iteration algorithm.

Value Iteration Algorithm

- Initialize $\vec{v}_0$
- For $t = 0, 1, 2, \dots, T-1$:

$$v_{t+1}(s) = \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v_t(s')]] \quad \forall s \in S$$

- Return $\vec{v}_T \xrightarrow{\text{green arrow}} \approx \vec{v}^*$

or

$$\pi_T(s) = \arg \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v_T(s')]] \quad \forall s \in S$$

$\vec{v}^{\pi_T} \xrightarrow{\text{green arrow}} \approx \vec{v}^*$

Bellman optimality operator: $\mathcal{T} : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$

$$(\mathcal{T} v)(s) = \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v(s')]] \quad \forall s \in S$$

Lemma: The Bellman optimality operator is a contraction mapping on $(\mathbb{R}^{|S|}, \|\cdot\|_\infty)$, i.e.,

$$\|\tau v' - \tau v\|_\infty \leq \gamma \|v' - v\|_\infty \quad \forall v, v' \in \mathbb{R}^{|S|}.$$

Proof:

$$\begin{aligned} \forall s \in S: |\tau v'(s) - \tau v(s)| &= \left| \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [v'(s')]] - \right. \\ &\quad \left. \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [v(s')]] \right| \\ &\leq \max_{a \in A} \left| R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [v'(s')] - \right. \\ &\quad \left. (R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [v(s')]) \right| \\ &= \max_{a \in A} \gamma \left| \mathbb{E}_{s' \sim P(\cdot|s, a)} [v'(s') - v(s')] \right| \\ &\leq \max_{a \in A} \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [|v'(s') - v(s')|] \\ &\leq \max_{s' \in S} \gamma |v'(s') - v(s')| \end{aligned}$$

$$\Rightarrow \max_{s \in S} |\tau v'(s) - \tau v(s)| \leq \gamma \max_{s \in S} |v'(s) - v(s)|$$

$$\Rightarrow \|\tau v' - \tau v\|_\infty \leq \gamma \|v' - v\|_\infty$$

■

Theorem: For any accuracy level $\epsilon > 0$ in the optimal value function,
 if we run the VI algorithm for $T \geq \frac{\log(\frac{\|v_0 - v^*\|_\infty}{\epsilon})}{\log \frac{1}{\gamma}}$,
 then, $\|v_T - v^*\|_\infty \leq \epsilon$.

Proof: Based on the previous lemma, after T iterations

$$\begin{aligned} \|v_T - v^*\|_\infty &= \|\tau v_{T-1} - \tau v^*\|_\infty \\ &\leq \gamma \|v_{T-1} - v^*\|_\infty \\ &= \gamma \|\tau v_{T-2} - \tau v^*\|_\infty \\ &\leq \gamma^2 \|v_{T-2} - v^*\|_\infty \\ &\vdots \\ &\leq \gamma^T \|v_0 - v^*\|_\infty \end{aligned}$$

we want $\|v_T - v^*\|_\infty \leq \epsilon$. It will hold if $T \geq \frac{\log(\frac{\|v_0 - v^*\|_\infty}{\epsilon})}{\log \frac{1}{\gamma}}$.

■

Note: The value of the final policy v^{π_T} returned by VI
 will not exactly correspond to the final value function v^* .

Theorem: The value of the final policy v^{π_T} returned by the VI algorithm satisfies $v^*(s) - v^{\pi_T}(s) \leq \frac{2\gamma^{\tau+1}}{1-\gamma} \|v_0 - v^*\|_\infty \quad \forall s \in S$.

Proof:

step 1

We want to show that for all $s \in S$ and for all $t \in \mathbb{N}_0$:

$$v^{\pi_t}(s) - v^*(s) \geq \gamma \mathbb{E}_{s' \sim P(s, \pi_t(s))} [v^{\pi_t}(s') - v^*(s')] - 2\gamma^{t+1} \|v_0 - v^*\|_\infty \quad (1)$$

$$\text{Let } \pi_t = \arg \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v_t(s')]]$$

policy π_t based on value function v_t

$$v^{\pi_t}(s) - v^*(s) \stackrel{\text{Bellman consistency equation for } v^{\pi_t}}{=} R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi_t(s))} [v^{\pi_t}(s')] - v^*(s)$$

$$\text{add and subtract two terms} \leftarrow = R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s')] - v^*(s)$$

$$+ \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')]$$

$$+ \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^t(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^t(s')]$$

$$\text{definition of } \pi_t \leftarrow \geq R(s, \pi_t^*(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t^*(s))} [v^t(s')]$$

+ rearranging terms

$$+ \gamma \mathbb{E}_{s' \sim P(s, \pi_t(s))} [v^{\pi_t}(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')] - v^*(s)$$

$$+ \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^t(s')]$$

Bellman consistency equation for v^*

$$\begin{aligned}
 &= -\gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t^*(s))} [v^*(s')] + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t^*(s))} [v^t(s')] \\
 &\quad + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')] \\
 &\quad + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^t(s')]
 \end{aligned}$$

Linearity of expectation

$$\begin{aligned}
 &= \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t^*(s))} [v^t(s') - v^*(s')] \\
 &\quad + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s') - v^*(s')] \\
 &\quad + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^*(s') - v^t(s')]
 \end{aligned}$$

avg $\geq$ -max(abs)

$$\geq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s') - v^*(s')] - 2\gamma \|v^* - v^t\|_\infty$$

⋮

repeat the argument

$$\begin{aligned}
 &\geq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s') - v^*(s')] \\
 &\quad - 2\gamma^{t+1} \|v^* - v_0\|_\infty
 \end{aligned}$$

step 2

$$v^*(s) - v^{\pi_t}(s) \stackrel{\textcircled{1}}{\leq} 2\gamma^{t+1} \|v_0 - v^*\|_\infty - \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s') - v^*(s')]$$

$$\stackrel{\textcircled{1}}{\leq} 2\gamma^{t+1} \|v_0 - v^*\|_\infty + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [2\gamma^{t+1} \|v_0 - v^*\|_\infty - \gamma \mathbb{E}_{s'' \sim P(\cdot | s', \pi_t(s'))} [v^{\pi_t}(s'') - v^*(s'')]]$$

constant

linearity of expectation
+
law of total expectation

$$= 2\gamma^{t+1} \|v_0 - v^*\|_\infty$$

$$+ 2\gamma^{t+2} \|v_0 - v^*\|_\infty$$

$$- \gamma^2 \mathbb{E}_{s''} [v^{\pi_t}(s'') - v^*(s'')]$$

repeat the argument

$$\vdots \leq 2\gamma^{t+1} \underbrace{\sum_{i=0}^k \gamma^i}_{\text{geometric series}} \underbrace{\|v_0 - v^*\|_\infty}_{\text{constant}}$$

$$- \gamma^k \mathbb{E}_{s_k} [v^{\pi_t}(s_k) - v^*(s_k)]$$

bounded if $R(\cdot, \cdot)$ bounded & $\gamma < 1$

let $k \rightarrow \infty$ $\gamma < 1 \Rightarrow v^*(s) - v^{\pi_t}(s) \leq \frac{2\gamma^{t+1}}{1-\gamma} \|v_0 - v^*\|_\infty$ (2)

